

NGUYEN VAN HUNG

Faculty of Mathematics and Informatics, Hanoi University of Science and Technology (HUST)

hungalberto2805@gmail.com ♦ Hung.NV227121@sis.hust.edu.vn ♦ (+84) 368 700 645

[GitHub](#) ♦ [LinkedIn](#) ♦ Hanoi, Vietnam

SUMMARY

Applied Mathematics and Informatics student at Hanoi University of Science and Technology with experience in mathematical optimization, scientific computing, machine learning, and LLM/RAG systems. Experienced in developing algorithms from theoretical analysis and convergence proofs to Python implementation and real-world experimentation. Interested in mathematically grounded AI and research-driven solutions to practical problems.

EDUCATION

Hanoi University of Science and Technology (HUST) | CPA: 3.55/4.00 Oct 2022 – Present

Honors and awards:

- Excellence Scholarship, Type A — Semester 2022.1 (*Top 1 GPA, cohort-wide*)
- Outstanding Student Award 2025 (*Sinh vien 5 tot*), HUST
- Certificate of Merit, Faculty of Mathematics and Informatics (2022–2023, 2023–2024)
- Certificate of Merit from the President of HUST for academic achievement and contributions to student organizations (2023–2024)
- First Prize, “*Market Economy in Vietnam and the World*” Essay Competition, HUST

Vingroup Applied AI Talent Program – VinUniversity | Cohort 3 Jul 2026 – Present

Participant

- Participating in an intensive applied AI training program focused on artificial intelligence foundations, practical AI development, and real-world problem solving. Receiving hands-on training through technical projects and practical business problems under the guidance of VinUniversity faculty and industry mentors.

RESEARCH ACTIVITIES

Student Research Conference – HUST

May 2026

Oral Presentation

- “*A Self-Adaptive Inertial CQ Algorithm with Conjugate-Gradient Direction for Split Feasibility Problems: Applications to Sparse Recovery and Elastic Net*”. The results of this study have been published in *Thang Long Journal of Science – Mathematics and Mathematical Sciences*, Vol. 5, No. 1 (2026). [[Paper](#)]

ACADEMIC PROJECTS

An Accelerated Inertial Relaxed Algorithm with a Conjugate-Gradient Direction for Split Feasibility Problems: Applications to Sparse Recovery and Extreme Learning Machines

2025 – 2026

Graduation Thesis – Faculty of Mathematics and Informatics, HUST | [[PDF](#)] [[Code](#)]

- Developed a self-adaptive inertial relaxed projection algorithm with a conjugate-gradient-type correction direction for split feasibility problems; established strong convergence to the minimum-norm solution and implemented the method in Python.
- Evaluated the method on numerical benchmarks, high-dimensional sparse signal recovery, and Extreme Learning Machine training for breast cancer classification, demonstrating strong performance against recent algorithms.

LitReview Agent – AI-Powered Systematic Literature Review System

2026

Team Project – Build Phase, VinUniversity

- Developed an AI-powered system to automate the systematic literature review workflow based on PRISMA, PICO, including research-scope refinement, criteria generation, academic search, paper verification, deduplication, and full-text RAG over scientific PDFs.
- Built the LLM/RAG pipeline using Python, FastAPI, LangGraph/LangChain, PostgreSQL/pgvector and ChromaDB, integrating Google Scholar/OpenAlex for paper discovery and supporting grounded document Q&A and export to BibTeX/Excel.

Improved Bregman Iteration for Sparse Portfolio Optimization

2024 – 2025

Capstone Project II – Faculty of Mathematics and Informatics, HUST

- Formulated a Markowitz mean-variance model with ℓ_1 regularization for sparse portfolio selection and developed an improved Bregman iterative algorithm combined with FISTA.
- Established convergence properties, implemented the solver in Python, and validated it on real market data, demonstrating improved sparsity, stability, and risk control over classical benchmarks.

E-Commerce Data Warehouse and Business Intelligence for Brazilian Olist

2025 – 2026

Course Project – Data Warehouse and Business Intelligence, HUST

- Built an end-to-end data warehouse pipeline for the 9-table Brazilian Olist e-commerce dataset (~126 MB) using Python and PostgreSQL, including data exploration, cleaning, transformation, feature engineering, and an OLAP model with 5 dimension tables and 3 fact tables.
- Developed Power BI dashboards for revenue, orders, and customer reviews, and extended the analysis with RFM, K-Means customer segmentation, and sales forecasting using SARIMA, Prophet, and XGBoost.

Temporal Kolmogorov–Arnold Transformer for Multivariate Time-Series Forecasting

2024

Course Project – Time Series Analysis, HUST

- Investigated the mathematical foundations of Kolmogorov–Arnold Networks (KAN), grounded in the Kolmogorov–Arnold representation theorem for multivariate functions.
- Implemented TKAT, integrating KAN layers into a Temporal Fusion Transformer (TFT) architecture; benchmarked on multiple forecasting datasets.

ACADEMIC INVOLVEMENT

Volunteer Team, Faculty of Mathematics and Informatics – HUST

Nov 2023 – Nov 2024

Captain

- Led planning and execution of large-scale volunteer campaigns (*Green Summer, Warm Winter*).
- Designed and delivered educational activities for children in remote mountainous regions.

Applied Mathematics & Informatics Club (AMIC) — HUST

Oct 2023 – Present

Member, Academic Committee

- Co-organized seminars, workshops, and competitions in applied mathematics and mathematical modeling.

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, Pandas, PyTorch, Matplotlib), SQL, Git/GitHub
Modeling	Machine learning, deep learning, time-series modeling, LLM/RAG, scientific computing
Mathematical	Optimization, numerical methods, iterative algorithms
Languages	Vietnamese (native), English (IELTS 6.0, TOEIC 805)